Biology

Topic: Anatomy of flowering plants

The Tissues

1. The tissue is a group of cells that
(A) Have a common origin.
(B) Are always structurally similar.
(C) Usually perform a common function.
1) Only (A) is correct.
2) (B) and (C) are correct
3) (A) and (C) are correct.
4) (A), (B), and (C) are correct.

2. Read the following statements and choose the incorrect option.
1) A plant is made up of different kinds of tissues.
2) Tissues are classified into two main groups, namely, simple and permanent tissues.
3) Different organs in a plant show differences in their internal structure.
4) Within angiosperms, the monocots and dicots are also seen to be anatomically different.

3. During the formation of leaves and elongation of stem, some cells are left behind from the shoot apical meristem which constitute the
1) Lateral meristem
2) Axillary bud
3) Intercalary meristem
4) Apical bud

4. A branch or a flower is developed in the axil of leaves by
1) Axillary bud
2) Intercalary meristem
3) Shoot apical meristem
4) Accessory bud

5. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): Intercalary meristem occurs between mature tissues.
Reason (R): Intercalary meristem helps in the elongation of the organ.
In the light of the above statements, choose the most appropriate answer from the options given below:
1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
2) (A) is correct but (R) is not correct.
3) (A) is not correct but (R) is correct.
4) Both (A) are (R) are correct and (R) is the correct explanation of (A).

6. Which of the following features given below are exhibited by primary meristems (A) and secondary meristems (B)?

(i) Develop at the later stage of life by dedifferentiation of permanent tissue.

(ii) Appear early in the life of plant.

(iii) Cells found in the active state of division.

(iv) Their activity results in increase in the diameter of plant organ.

	A	B
1)	(ii), (iii), (iv)	(i)
2)	(ii), (iii)	(i), (iv)
3)	(i), (ii)	(iii), (iv)
4)	(ii), (iv)	(i), (iii)

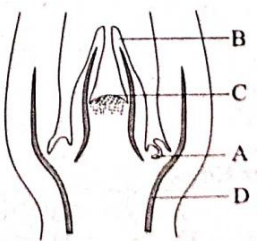
7. Plants tissues are classified into two main groups, i.e., meristematic and permanent tissues mainly based on

1) Cell's structure and function
2) The cells being formed are capable of growing or not
3) The cells being formed are capable of dividing or not
4) All of the above

8. Secondary meristem is all except

1) Cork cambium
2) Intrafascicular cambium
3) Interfascicular cambium
4) None of the above

9. Examine the figure given below and select the correct option that represents the correct labelling A, B, C and D.



	A	B	C	D
1)	Leaf primordium	Meristematic zone	Axillary bud	Intercalary meristem
2)	Leaf primordium	Short optical meristem	Axillary bud	Cork cambium
3)	Axillary bud	Leaf primordium	Shoot apical meristem	Differentiating vascular tissue
4)	Axillary bud	Shoot apical meristem	Meristematic zone	Leaf primordia

10. Following divisions of cells in both primary as well as secondary meristems, the newly formed cells become structurally and functionally specialised and lose the ability to divide. Such cells are termed as

1) Mature cells
2) Permanent cells

3) Secondary cells
4) Both (1) and (2)

11. Select the correct statement with respect to meristem.
1) It is a group of large, isodiametric mature cells.
2) It is the area of active cell division in plants.
3) Cells possess thin cytoplasm and small nucleus.
4) Cells show very low metabolic activities.

12. The tissue that occurs in grasses and regenerates parts removed by grazing herbivores is
1) Axillary meristem
2) Apical meristem
3) Intercalary meristem
4) Lateral meristem

13. The meristems that occurs in the mature regions of roots and shoots of many plants, particularly those that produce woody axis are called
1) Secondary meristem
2) Apical meristem
3) Intercalary meristem
4) Root and shoot apical meristem respectively

14. A simple permanent tissue
1) Is made up of more than one type of cells that work together as a unit.
2) Is composed of cells that are similar in structure but differ in function.
3) Is made up of only one type of cells and perform a common function.
4) Is composed of cells that are structurally similar but differ in function.

15. Identify the tissue from the given features.
I. The cells of this tissue are living.
II. The cell show angular wall thickening.
III. The cells are also instrumental in providing mechanical support.
1) Parenchyma
2) Collenchyma
3) Sclerenchyma
4) Xylem

The Tissue System

16. Based on the structure and _A_, there are _B_types of tissue system in plant body. Choose the correct option to fill in the blanks.

	A	**B**
1)	Function	Three
2)	Location	Three
3)	Function	Four
4)	Location	Two

17. The epidermal tissue system is characterised by all of these, except

1) Parenchymatous cells without intercellular space.
2) Unicellular root hairs.
3) Being always multilayered.
4) Trichomes in stem epidermis.

18. Epidermis is absent in which of the following pairs of parts of a flowering plant?
1) Style and stigma
2) Root tip and shoot tip
3) Petiole and pedicel
4) Shoot bud and floral bud

19. Read the following statements carefully and select the correct option.
A. Guard cells are living, have nucleus and chloroplasts.
B. The outer wall of guard cells is thick and elastic.
C. The stomatal aperture, guard cells and the surrounding subsidiary cells are together called the stomatal apparatus.
1) Only (A)
2) (B) and (C)
3) (A) and (C)
4) (A), (B), and (C)

20. In grasses, guard cells of stomata are
1) Kidney shaped
2) Dumb-bell shaped
3) Sickle shaped
4) Bean shaped

21. Match the following columns and choose the correct option.

Column I		**Column II**	
A	Bulliform cells	i	Waxy layer
B	Stomata	ii	Single layer
C	Cuticle	iii	Guard cells
D	Epidermis	iv	Empty colourless cell

1) A-i, B-ii, C-iii, D-iv
2) A-iv, B-iii, C-i, D-ii
3) A-iii, B-ii, C-i, D-iv
4) A-ii, B-iii, C-i, D-iv

22. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): In dicotyledonous stems, the vascular bundles are considered as open.
Reason (R): The vascular bundles in dicots have no cambium present in them.
In the light of the above statements, choose the most appropriate answer from the options given below:
1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
2) (A) is correct but (R) is not correct.
3) (A) is not correct but (R) is correct.

4) Both (A) are (R) are correct and (R) is the correct explanation of (A).

23. Sometimes a few epidermal cells in the vicinity of guard cells become specialised in their shape and size, which are known as

1) Subsidiary cells
2) Complimentary cells
3) Mesophyll cell
4) Conjunctive cells

24. In leaves, ground tissue consists of

1) Mesophyll
2) Palisade tissue only
3) Spongy parenchyma only
4) Epidermis and bundle sheath

25. Parenchymatous cells are present in

1) Cortex and pericycle
2) Medullary rays
3) Pith
4) All of these

26. Given below are two statements:

Statement I: The conjoint vascular bundles usually have the phloem located only on the outer side of xylem.

Statement II: In conjoint type of vascular bundles, the xylem and phloem are jointly situated along the same radius of vascular bundles

In the light of the above statements, choose the most appropriate answer from the options given below:

1) Both statement I and II are incorrect.
2) Statement I is correct, but statement II is incorrect.
3) Statement I is incorrect, and statement II is correct.
4) Both statement I and II are correct.

27. Ground tissue includes

1) All the tissues internal to endodermis
2) All tissues external to endodermis
3) All tissues except epidermis and vascular tissues
4) All tissues except epidermis and endodermis

28. What among the following is the epiblema of roots equivalent to?

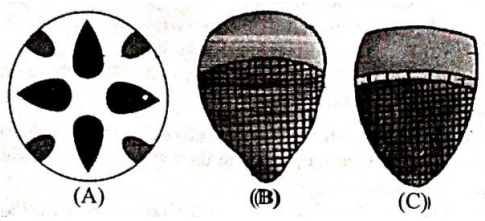
1) Epidermis
2) Endodermis
3) Ectodermis
4) Stele

29. The ground tissue system is made up of

1) Epidermis and vascular tissues
2) Simple and complex permanent tissues
3) Simple tissues such as parenchyma, collenchyma, and sclerenchyma

4) Only thin-walled simple permanent tissue

30. The types of vascular bundles (A, B and C) given below are present in



	A	B	C
1)	Monocot leaf	Dicot stem	Monocot stem
2)	Root	Dicot stem	Monocot stem
3)	Root	Monocot stem	Dicot stem
4)	Monocot root	Monocot leaf	Monocot stem

31. (i) Xylem and phloem are present in alternating separate radii. Such vascular bundles are called _A_.

(ii) Xylem and phloem are present on same radius. Such vascular bundles are called_B_.

Choose the correct option to fill in the blanks.

	A	B
1)	Radial	Conjoint
2)	Radial	Concentric
3)	Conjoint	Radial
4)	Conjoint	Concentric

32. We remove _____ when we peel the skin of a potato tuber

Choose the correct option to fill in the blank.

1) Endodermis
2) Epidermis
3) Cuticle
4) Periderm

33. Which of the following is not a part of stele?

1) Endodermis
2) Pith
3) Pericycle
4) Vascular tissue

34. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): In the monocotyledons, the vascular bundles are referred to as closed.

Reason (R): Phloem is located only on the inner side of xylem.

In the light of the above statements, choose the most appropriate answer from the options given below:

1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
2) (A) is correct but (R) is not correct.
3) (A) is not correct but (R) is correct.

4) Both (A) are (R) are correct and (R) is the correct explanation of (A).

35. Match the following columns and choose the correct option.

Column I		**Column II**	
a	Radial	i	Dicots
b	Conjoint open	ii	Roots
c	Conjoint closed	iii	Monocots
d	Guard cells	iv	Can be branched or unbranched
e	Trichomes	v	Enclose stomatal pore

1) a - ii, b – iv, c – i, d-v, e- iii
2) a-ii, b – i, c – iii, d - v, e – iv
3) a- iii, b – i, c – ii, d – iv, e-v
4) a - iii, b - ii, c - i, d - iv, e - v

36. Read the following statements:

A. Epidermal cells are parenchymatous cells and have a small amount of cytoplasm and a large vacuole.

B. Waxy layer cuticle is absent in roots.

C. The xylem and phloem together constitute the vascular bundles.

Find correct option.

1) Only (B) and (C) are correct.
2) Only (C) is correct.
3) Only (B) is correct.
4) (A), (B), and (C) are correct.

Anatomy of Dicotyledonous and Monocotyledonous Plants

37. In the transverse section of _____ , one can find a conjoint and open vascular bundle.

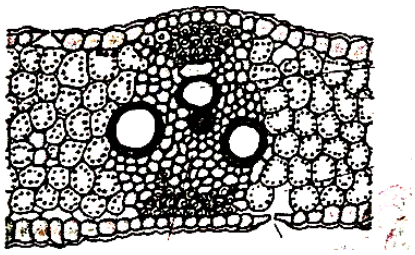
1) Dicot stem
2) Dicot root
3) Monocot stem
4) Monocot root

38. Match the following columns and choose the correct option.

Column I		**Column II**	
1	Endodermis	a	Suberin
2	Casparian strips	b	Barrel shaped
3	Pericycle	c	Lies between the xylem and the phloem
4	Conjuctive tissue	d	Thick-walled parenchymatous cells

1) 1-a, 2-b, 3-d, 4 – c
2) 1-b, 2-a, 3-d, 4-c
3) 1-b, 2-a, 3-c, 4-d
4) 1-a, 2-b, 3-c, 4-d

39. Identify the given figure and select the incorrect statement related to it.



1) Stomata present on both leaf surfaces
2) Differentiated leaf mesophyll
3) Presence of parallel venation
4) Certain adaxial epidermal cells modify to form bulliform cells

40. Given below are two statements:

Statement I: In grasses, certain adaxial epidermal cells along with the veins modify themselves into large empty, colourless cells, called bulliform cells.

Statement II: All tissues on the outer side of endodermis such as pericycle, vascular bundles and pith constitute the stele in dicot root.

In the light of the above statements, choose the most appropriate answer from the options given below:

1) Both statement I and statement II are correct.
2) Both statement I and statement II are incorrect.
3) Statement I is correct, but statement II is incorrect.
4) Statement I is incorrect, but statement II is correct.

41. What would be the colour of the stained xylem and phloem if a transverse section of a stem is first stained with safranin and later with fast green followed by usual schedule of double staining for the preparation of a permanent slide?

1) Orange and Blue
2) Red and Green
3) Orange and Green
4) Red and Yellow

42. Hypodermis is collenchymatous for mechanical strength in

1) Dicot root
2) Monocot root
3) Dicot stem
4) Monocot stem

43. The condition where the first formed xylem element faces the pericycle is called A and it is seen in B. Choose the correct option to fill in the blanks.

	A	**B**
1)	Exarch	Root
2)	Endarch	Root
3)	Exarch	Stem
4)	Mexarch	Root

44. Given below are certain characteristic features.

i. Unicellular hair
ii. Endodermis with barrel-shaped cells

iii. Pith-small/inconspicuous
iv. Radiial vascular bundle
v. Xylem exarch
vi. 2-4 xylem and phloem

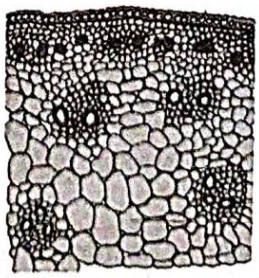
The above features refers to which of the following?

1) Monocot root
2) Dicot root
3) Monocot stem
4) Dicot stem

45. Find the correctly matched pair.

1) Casparian strip-Root endodermis
2) Well-developed pith-Dicot stem
3) Closed vascular bundle-Cambium absent
4) All of the above

46. Identify the given figure and select the incorrect option.



1) Collenchymatous hypodermis
2) Small peripheral vascular bundles
3) Parenchymatous ground tissue
4) Vascular bundles conjoint and closed

47. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): The anatomy of the monocot root is similar to the dicot root in many aspects.

Reason (R): Both dicot root and monocot root have polyarch xylem bundles.

In the light of the above statements, choose the most appropriate answer from the options given below:

1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
2) (A) is correct but (R) is not correct.
3) (A) is not correct but (R) is correct.
4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

48. Match the following columns and choose the correct option.

Column I		**Column II**	
1	Isobilateral leaf	a	Sclerenchymatous cell
2	Dorsiventral leaf	b	More stomata are present on abaxial epidermis
3	Bundle sheath cell	c	Parenchymatous cells
4	Cortex	d	Stomata are present on both the surfaces of

			the epidermis

1) 1-d, 2-b, 3- a, 4-c
2) 1-d, 2-b, 3-c, 4-a
3) 1-b, 2-d, 3- a, 4- c
4) 1-b, 2-d, 3-c, 4-a

49. The following anatomical features are related with

(i) Presence of sclerenchymatous hypodermis.
(ii) Presence of water-containing cavities within vascular bundles.
(iii) Large number of scattered vascular bundles.
(iv) Vascular bundles are conjoint and closed.

1) Maize stem
2) Wheat root
3) Mustard stem
4) Cucurbita root

50. Select the correct order of layers of cells from centre to periphery in dicot stem:-

1) Epidermis→ Endodermis → Hypodermis → General cortex → pericycle
2) Epidermis → Hypodermis→ General cortex→ Endodermis → pericycle
3) Pericycle → Endodermis → General cortex → Hypodermis → Epidermis
4) Pericycle → Endodermis →→ Hypodermis → General cortex → Epidermis

51. In the vascular bundle of dorsiventral leaf, xylem lies towards the _A_ and phloem lies towards the_B_

Choose the correct option to fill in the blanks.

	A	**B**
1)	Adaxial epidermis	Palisade tissue
2)	Adaxial epidermis	Abaxial epidermis
3)	Abaxial epidermis	Palisade tissue
4)	Spongy parenchyma	Abaxial epidermis

52. Match the following columns and choose the correct option

Column I		**Column II**	
1	Trichomes	a	Specialised epidermalcells
2	Root hairs	b	Absent in roots
3	Subsidiary cells	c	Usually multicellular
4	Cuticle	d	Always unicellular

1) 1-c, 2-d, 3-a, 4-b
2) 1-c, 2-d, 3-b, 4-a
3) 1-d, 2-c, 3-a, 4-b
4) 1-a, 2-b, 3- c, 4- d

53. Read the following statements with respect to dicot stem and select the correct option.

A. Epidermis is covered with a thin layer of cuticle, may bear trichomes and few stomata.
B. Ring arrangement of conjoint, collateral, and open vascular bundles.
C. Presence of semilunar patches of sclerenchyma above the phloem.

D. Presence of conjunctive tissue between the xylem and the phloem.

1) Only statement (A) is correct.
2) Statements (B), (C), and (D) are correct,
3) Statements (A), (B), (C), and (D) are correct.
4) Only statement (D) is incorrect.

54. Dicot stem differs from monocot stem in lacking all, except

1) Bundle sheath cells
2) Pith and medullary rays
3) Protoxylem cavity
4) Scattered vascular bundles

Secondary Growth

55. Secondary growth occurs due to the activity of

1) Vascular cambium
2) Only interfascicular cambium
3) Phellogen
4) Both (1) and (3)

56. Secondary growth usually does not occur in

1) Stem and root of dicots.
2) Stem and root of monocots.
3) Stem and root of gymnosperms.
4) More than one option is correct.

57. In dicot stem, secondary growth in extra stelar region is due to

1) Interfascicular cambium
2) Intrafascicular cambium
3) Intercalary meristem
4) Cork cambium

58. State true (T) or false (F) for the given statements and select the correct option.

(A) All tissue layers exterior to vascular cambium constitute bark.
(B) Phellogen, phellem and phelloderm are collectively known as periderm.
(C) The peripheral region of secondary xylem is lighter in colour and is known as heartwood.
(D) Lenticels are lens-shaped openings for gaseous exchange

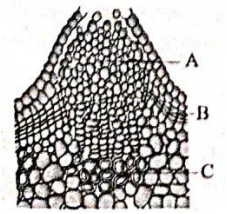
	A	B	C	D
1)	T	T	F	T
2)	T	F	F	F
3)	F	F	T	T
4)	F	T	T	F

59. Which is not true about phellogen?

1) It is extra stelar cambium.
2) It develops from differentiation of cortical cells.
3) It is made up of narrow, thin-walled nearly rectangular cells.

4) It is also known as cork cambium.

60. Examine the figure given below and select the correct labels for A, B and C.



	A	B	C
1)	Complementary cells	Phellogen	Phellem
2)	Phellem	Cork cambium	Periderm
3)	Complementary cells	Cork cambium	Secondary cortex
4)	Phellem	Phellogen	Phelloderm

61. Which ofthe following is an incorrect difference between sapwood and heartwood?

Column I		Column II	
1)	Forms the peripheral major portion	1)	Forms the central major portion
2)	Lighter in colour	2)	Darker in colour
3)	Less durable and susceptible to attack of microorganisms and insects	3)	Hard, more durable, and resistant to attack of pathogens
4)	Physiologically inactive and cannot conduct water	4)	Physiologically active and involved in the conduction of water

62. During secondary growth in dicot stem, at certain regions cork cambium cuts off parenchymatous cell on the outer side instead of _____

1) Cork cells
2) Phelloderm
3) Phellogen
4) Secondary phloem

63. What leads to the formation of interfascicular cambium and cork cambium?

1) Cell division
2) Cell dedifferentiation
3) Redifferentiation
4) Cell differentiation

64. What do phellogen and phellem denote respectively?

1) Cork and secondary cortex
2) Secondary cortex and cork
3) Cork and cork cambium
4) Cork cambium and cork

65. Cortex that develops during the secondary growth of dicot stem is

1) Secondary cortex

2) Phelloderm
3) General cortex
4) Both (1) and (2)

66. Secondary phloem is nearer to
1) Secondary xylem
2) Cork cambium
3) Vascular cambium
4) Primary cortex

67. Read the following statements and mark them as true (T) or false (F).
A. Intrafascicular cambium develops in between secondary xylem and secondary phloem during the secondary growth of plant organ.
B. Interfascicular cambium develops from cells of medullary rays by means of dedifferentiation.
C. The vascular cambium is generally more active on the inner side.
D. The vascular cambium ring cuts off secondary xylem away from the centre.

	A	B	C	D
1)	T	T	T	F
2)	F	T	T	F
3)	F	T	T	T
4)	T	F	F	T

68. All of these are formed through dedifferentiation, except
1) Cork cambium
2) Interfascicular cambium
3) Intrafascicular cambium
4) None of the above

69. Bark includes
1) All tissues exterior to vascular cambium
2) Periderm and secondary phloem
3) Both living and dead tissues
4) All of these

70. At places, the cork contains aerating pores called
1) Stomata
2) Lenticels
3) Hydathode
4) Pneumatothodes

71. Secondary growth in dicot roots is not characterised by
1) Wavy ring of vascular cambium
2) Dedifferentiation of cells just below the phloem bundles.
3) Vascular cambium being half primary and half secondary in origin.
4) Purely secondary origin of cork cambium.

72. How many of the given structures are redifferentiated structures developed during secondary growth?

Interfascicular cambium, secondary xylem, secondary phloem,

phellogen, cork, primary phloem, secondary cortex, complementary cells

1) Five
2) Six
3) Four
4) Seven

73. Given below are four statements (A-D) regarding secondary growth in dicot root.

A. The vascular cambium is completely secondary inorigin.
B. Vascular cambium develops from the tissue located first below the phloem bundle and a portion of pericycle tissue.
C. Annual rings are clearly distinct.
D. Phellogen cuts phellem on the inner side and phelloderm towards the outer side.

Which two statements are incorrect?

1) (A) and (D)
2) (A) and (C)
3) (B) and (C)
4) (C) and (D)

74. Consider the following statements and select the option that correctly fill in the blanks.

I. In spring season, cambium is A active and produces B wood.
II. In autumn season, cambium is C active and produces D wood.
III. Bark formed early in the season is called E bark and towards the end of season is called F bark.

	A	**B**	**C**	**D**	**E**	**F**
1)	More	Early	Less	Late	Soft	Hard
2)	More	Late	Less	Early	Ring	Scaly
3)	Less	Early	More	Late	Soft	Hard
4)	Less	Late	More	Early	Hard	Soft

Answer Key

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
3	2	2	1	4	2	3	2	3	4	2	3	1	3	2
16	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
2	3	2	3	2	2	2	1	1	4	4	3	1	3	3
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**	**41**	**42**	**43**	**44**	**45**
1	4	1	2	2	2	4	1	2	2	3	2	3	1	2
46	**47**	**48**	**49**	**50**	**51**	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
4	1	2	1	1	3	2	4	4	2	4	2	4	1	2
61	**62**	**63**	**64**	**65**	**66**	**67**	**68**	**69**	**70**	**71**	**72**	**73**	**74**	**75**
3	4	1	2	4	4	3	2	3	4	2	3	1	4	1